

String-Stable Cooperative Adaptive Cruise Control With Minimized Time Headway in the Face of Delayed Communication

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Abstract—The cooperative adaptive cruise control problem is considered in a one-vehicle lookahead configuration with first-order feedback and feedforward controllers. The feedback controller uses the spacing error defined with a constant time gap, while the feedforward controller filters the preceding vehicle input (as the desired acceleration) received through delayed communication. The communication delay is assumed to be constant and yet unknown. A lower bound is then derived for the legitimate time headway choices in terms of the maximum possible communication delay and the vehicle time constant. The delay margin of the feedback loop turns out to be quite large with the sub-optimal controllers. On the other hand, a major drawback of approaching the infimum time headway is observed as the increased noise sensitivity of the feedback loop especially in the case of small maximum communication delays.

Index Terms—Cooperative control, delay systems, robust control.

I. INTRODUCTION

THE COOPERATIVE adaptive cruise control (CACC) emerged as an upgrade of adaptive cruise control (ACC) in parallel with the developments in wireless communications [3], [8], [13], [14], [15], [16], [17]. Though both systems have the same goal of maintaining a certain distance to the preceding vehicle, ACC relies solely on onboard sensors, while CACC facilitates the use of information received through communication. As a result, CACC systems offer potential improvements in driver comfort, safety and fuel consumption. A platoon of connected and automated vehicles would help increase traffic capacity and improve fuel economy, provided that the underlying CACC system is designed properly.

Suitable design of (C)ACC systems is formulated by imposing a string stability requirement. This typically reads as the avoidance of amplification of certain motion signals (like spacing error, acceleration, control input etc.) in a preferred norm (backward) along the platoon (see [9] and the references therein). It is particularly convenient to work with the

$\mathcal{L}_2$ string stability condition as this leads to an $\mathcal{H}_\infty$ norm constraint on the transfer function that shapes the signal propagation. Such a condition could not be ensured in ACC while trying to maintain a constant distance to the preceding vehicle. Hence a viable alternative was proposed as the constant time gap (headway) policy, in which the desired inter-vehicle distance is increased with a term that is proportional to the velocity of the ego vehicle [10].

When applying the constant time gap policy, the time headway needs to be chosen sufficiently large to ensure string stability. With the transfer from desired to actual acceleration described by a first-order system identified by a time constant and the desired acceleration generated by a PD-type controller, [20] showed that the time headway has to be at least twice the time constant to ensure string stability. When an actuator delay is added to the system model, the required minimum time gap (with this particular PD-type controller) increases by twice the maximum delay [22]. Time headway requirements that emerge from the presence of a double integrator in the vehicle model are derived in a more generic form in [11]. Increased time gap would increase the inter-vehicle distances, thereby leading to an undesirable outcome of decreased traffic capacity. CACC made it possible to ensure string stability with smaller time gaps if compared to ACC [4], [6]. In fact with certain information flow topologies like leader and predecessor following, string stability can be achieved even with a constant spacing policy [19], [21]. Nevertheless, it is more sensible to consider CACC with a nonzero time gap since platoons will operate with typically delayed, noise-corrupted and occasionally dropped communication, which would evidently be detrimental to string stability.

Due to the natural concerns about traffic capacity, the minimum employable value of the time gap would be of strong interest in CACC. This needs to be determined based on the worst-case scenarios about communication impairments. A generic approach in the case of uncertain (yet constant) communication delay (with known maximum value) would be to employ robust control methods within a deterministic setting, by characterizing the delay uncertainty with a weighting filter [23]. One would, though, need to perform numerous optimizations to determine the minimum employable time gaps for different cases. Moreover, the resulting controllers would typically have high orders, further adjustment of which will not be easy. Simple controllers (e.g., of the sort in [20]) might hence

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be more appealing if it is known how to choose their parameters in a way to reduce the time gap as much as possible while still maintaining string stability in the face of communication delay. Investigations based on vehicle models that capture essential dynamics would then provide some insight into the fundamental limitations on minimum inter-vehicle spacings. An investigation in this direction is found in [2] with various feedforward strategies for heterogeneous platoons, but the results are obtained through extensive numerical optimization rather than analytical derivations. A similar feedback controller is considered by [5] together with acceleration feedforward in a heterogeneous platoon and a lower bound is derived on the time headway in terms of the feedback controller parameters (without going further to find the infimal value). The recent work [1] presents minimal time headway choices for a multiple-predecessor following topology in terms of the vehicle time constant, the communication delay and the relative acceleration feedback gain. Finally [12] presents minimal time headway choices in a specific leader and predecessor following configuration for a PD-type controller.

This letter considers the one-vehicle lookahead CACC for a homogenous platoon with first-order feedback and feedforward controllers. The feedback controller only needs the distance to the preceding vehicle and the ego vehicle velocity, while the feedforward controller uses the control input signal received with some communication delay from the preceding vehicle. By considering a third-order vehicle model and a homogenous platoon, we first derive propagation equations for the control input and the spacing error in the next section. The problem is then formulated as the determination of a lower bound on the legitimate time headway choices for string stability as a function of the maximum communication delay. Section III presents the derivation of the main result, which provides an analytical answer to this question. To the best of the author's knowledge, such a result has not yet appeared in the literature for one-vehicle lookahead CACC. In particular, our controller does not require acceleration information, which is crucial for the recent and particularly relevant works [1], [5] especially for handling heterogeneous platoons. The section is concluded by an analysis of the error response as well as the robust stability and noise sensitivity of the feedback loop with the suboptimal controllers. Example simulation results are provided in Section IV before drawing some final remarks.

II. PROBLEM STATEMENT

In this section we establish the setup and then derive the platoon equations needed for a clear problem formulation.

A. Vehicle Model and Spacing Error

We consider a model expressed for the i th vehicle as

$$\begin{aligned}\dot{x}_i(t) &= v_i(t); & x_i(t) &: \text{position} \\ \dot{v}_i(t) &= a_i(t); & v_i(t) &: \text{velocity} \\ \tau \dot{a}_i(t) &= u_i(t) - a_i(t); & a_i(t) &: \text{acceleration}\end{aligned}\quad (1)$$

where $u_i(t)$ represents the desired acceleration command as the control input. The time constant τ is assumed to be identical for all vehicles in a homogenous platoon. A more realistic model would also include an input delay, which will be

addressed briefly after the derivation of the main result. The vehicle model is expressed in the Laplace domain as

$$\left. \begin{aligned} s\tilde{x}_i(s) - x_i(0) &= v_i(s) \\ s\tilde{v}_i(s) - v_i(0) &= a_i(s) \\ \tau s a_i(s) - \tau a(0) &= u_i(s) - a_i(s) \end{aligned} \right\} \Rightarrow \quad (2)$$

$$\tilde{x}_i(s) = \underbrace{\frac{1}{s^2(\tau s + 1)}}_{\mathcal{P}_0(s)} u_i(s) + \underbrace{\frac{\tau a_i(0)}{s^2(\tau s + 1)} + \frac{v_i(0)}{s^2} + \frac{x_i(0)}{s}}_{w_i(s)}.$$

The transfer function from u_i to x_i is thus identified as $\mathcal{P}_0(s)$ and w_i is attributed to nonzero initial conditions.

The plant model in a platoon control configuration is slightly modified in connection with the spacing policy. Safe adaptive cruise control motivates a constant time gap (headway) policy, which amounts to a desired inter-vehicle spacing that increases proportionally with increasing vehicle velocity. With the proportionality constant represented by h (for headway), the spacing error is then defined as

$$e_i(t) = \underbrace{x_{i-1}(t) - x_i(t) - l_i}_{\text{distance to preceding vehicle}} - h v_i(t) - (g_i - l_i). \quad (3)$$

Here g_i is identified as the desired standstill distance between the rear bumpers, which should be chosen sufficiently larger than the length of the i th vehicle for safety: $g_i > l_i$. In order to be able to represent the spacing error as the tracking error in a control configuration, we introduce a clean output signal

$$y_i(s) = \underbrace{(hs + 1)}_{\mathcal{H}^{-1}(s)} [\tilde{x}_i(s) - w_i(s)] = \underbrace{\frac{hs + 1}{s^2(\tau s + 1)}}_{\mathcal{P}_h(s)} u_i(s), \quad (4)$$

where the plant is identified as $\mathcal{P}_h = \mathcal{H}^{-1}\mathcal{P}_0$. The spacing error is then expressed in the Laplace domain by

$$\begin{aligned} e_i(s) &= \tilde{x}_{i-1}(s) - \underbrace{(hs + 1)\tilde{x}_i(s)}_{y_i(s) + \mathcal{H}^{-1}(s)w_i(s)} + h x_i(0) - \frac{g_i}{s} \\ &= \underbrace{\tilde{x}_{i-1}(s) - w_{i-1}(s) + d_i(s)}_{\tilde{r}_i(s)} - y_i(s), \end{aligned} \quad (5)$$

where d_i is introduced as a disturbance term that will help describe error and control input propagation:

$$\begin{aligned} d_i(s) &= w_{i-1}(s) - (hs + 1)w_i(s) + h x_i(0) - \frac{g_i}{s} \\ &= \frac{\tau [a_{i-1}(0) - (hs + 1)a_i(0)]}{s^2(\tau s + 1)} + \frac{v_{i-1}(0) - v_i(0)}{s^2} \\ &\quad + \frac{x_{i-1}(0) - x_i(0) - h v_i(0) - g_i}{s}. \end{aligned} \quad (6)$$

Indeed this disturbance will be identically zero when the initial accelerations are zero, the initial velocities are equal and the initial inter-vehicle spacing is equal to its desired value, i.e., $x_{i-1}(0) - x_i(0) = h v_i(0) + g_i$. We thus arrive at a model of the i th platoon section as in Figure 1.

B. Spacing Error and Control Input Propagation

A CACC system can be viewed as a combination of feedback and feedforward control as in Figure 1. The feedback controller $\mathcal{C}_h$ acts on the spacing error of the ego (i th) vehicle, which would be obtained as in (3) based on the radar/lidar

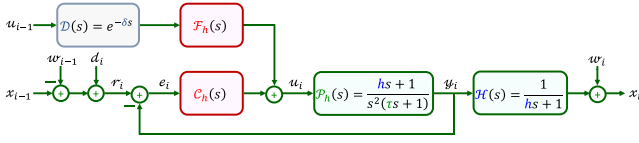


Fig. 1. Cooperative adaptive cruise control configuration.

measurement of the distance to the preceding vehicle and the measurement of the vehicle's own velocity. On the other hand, the feedforward controller $\mathcal{F}_h$ uses the control input of the preceding ($i-1$)th vehicle that will be received through (delayed) communication. The sensitivity and complementary sensitivity of the feedback loop are defined as

$$\mathcal{S}_h \triangleq \frac{1}{1 + \mathcal{P}_h \mathcal{C}_h} \text{ and } \mathcal{T}_h \triangleq \frac{\mathcal{P}_h \mathcal{C}_h}{1 + \mathcal{P}_h \mathcal{C}_h} = 1 - \mathcal{S}_h. \quad (7)$$

We now derive the propagation equations for the control input and the spacing error, by also including the disturbance effects for the sake of generality. From Figure 1, we infer

$$\begin{aligned} u_i &= \mathcal{F}_h \mathcal{D} u_{i-1} + \mathcal{C}_h (\bar{r}_i - \mathcal{P}_h u_i) \Rightarrow \\ u_i &= \mathcal{S}_h (\mathcal{F}_h \mathcal{D} u_{i-1} + \mathcal{C}_h \bar{r}_i), \end{aligned} \quad (8)$$

where $\mathcal{D}$ represents the transfer function of the communication channel. We next recall (5) and repeat the configuration of Figure 1 for the preceding vehicle to obtain

$$\bar{r}_i = \bar{x}_{i-1} - w_{i-1} + d_i = \mathcal{H} \mathcal{P}_h u_{i-1} + d_i. \quad (9)$$

By inserting this in (8), we arrive at

$$u_i = \mathcal{G} u_{i-1} + \mathcal{C}_h \mathcal{S}_h d_i, \quad (10)$$

where the transfer function that shapes the propagation is

$$\mathcal{G} \triangleq \mathcal{S}_h \mathcal{F}_h \mathcal{D} + \mathcal{T}_h \mathcal{H} = \mathcal{H} - \mathcal{S}_h (\mathcal{H} - \mathcal{F}_h \mathcal{D}). \quad (11)$$

Equations (9), (10) and (11) can now be used to derive

$$e_i = \bar{r}_i - \mathcal{P}_h u_i = \mathcal{M} u_{i-1} + \mathcal{S}_h d_i, \quad (12)$$

where the transfer from the control command of the preceding vehicle to the spacing error of the ego vehicle is

$$\mathcal{M} \triangleq \mathcal{P}_h (\mathcal{H} - \mathcal{G}) = \mathcal{S}_h \mathcal{P}_h (\mathcal{H} - \mathcal{F}_h \mathcal{D}). \quad (13)$$

C. Synthesis With Specific FB and FF Controllers

With the intention to derive analytical results and with inspirations from [15], [17], we now restrict interest to feedforward and feedback controllers as

$$\mathcal{F}_h = \underbrace{1}_{\mathcal{F}_0} \cdot \underbrace{\frac{1}{hs+1}}_{\mathcal{H}} = \frac{1}{\bar{h}\bar{s}+1}, \quad (14)$$

$$\mathcal{C}_h = \underbrace{\left(\frac{\kappa\phi}{\tau}s + \frac{\kappa}{\tau^2} \right)}_{\mathcal{C}_0} \cdot \underbrace{\frac{1}{hs+1}}_{\mathcal{H}} = \frac{\kappa}{\tau^2} \frac{\phi\bar{s}+1}{\bar{h}\bar{s}+1}. \quad (15)$$

Here we introduced a scaled Laplace variable as $\bar{s} \triangleq \tau s$ and a normalized time headway by

$$\bar{h} \triangleq \frac{h}{\tau} > 0. \quad (16)$$

The choice of the feedforward controller as in (14) would be ideal when communication is perfect (i.e., $\mathcal{D} = 1$). Indeed this would render $\mathcal{M} = 0$ and $\mathcal{G} = \mathcal{H}$, thus ensuring string stability as well as perfect response to preceding vehicle maneuvers. It would then be possible to choose h arbitrarily close to zero. On the other hand, the choice of the feedback controller as in (15) renders $\mathcal{S}_h = \mathcal{S}_0$ and $\mathcal{T}_h = \mathcal{T}_0 = 1 - \mathcal{S}_0$ with

$$\mathcal{S}_0(s) = \frac{1}{1 + \mathcal{P}_0(s)\mathcal{C}_0(s)} = \frac{\mathcal{N}_0(\bar{s})}{\mathcal{Q}_0(\bar{s})} = \frac{\bar{s}^2(\bar{s}+1)}{\bar{s}^3 + \bar{s}^2 + \kappa\phi\bar{s} + \kappa}. \quad (17)$$

The platoon behavior is shaped by the propagation equations in (10) and (12). Envisioning a different controller during the phase of joining a platoon, we disregard the disturbance terms in these equations and express the transfer functions of primary interest for us in terms of the specific controller ingredients as

$$\mathcal{G} = \mathcal{H}(1 - \mathcal{S}_0(1 - \mathcal{F}_0\mathcal{D})) = \mathcal{H} \cdot \left[1 - \frac{\mathcal{N}_0}{\mathcal{Q}_0}(1 - \mathcal{D}) \right], \quad (18)$$

$$\mathcal{M} = \mathcal{P}_0 \mathcal{S}_0(1 - \mathcal{F}_0\mathcal{D}) = \frac{\tau^2(1 - \mathcal{D})}{\mathcal{Q}_0}. \quad (19)$$

We are interested in analysis and synthesis in the face of uncertain (yet constant) communication delay. The channel is hence represented by a delay transfer function with a normalized bound for the uncertain delay as follows:

$$\mathcal{D} = e^{-\delta s}, \text{ where } \delta \in [0, \bar{\delta}\tau] \text{ with a known } \bar{\delta}. \quad (20)$$

Our goal is to derive an analytical result that relates the legitimate choices of the normalized time headway $\bar{h}$ to the maximum normalized communication delay $\bar{\delta}$.

The $\mathcal{L}_2$ string stability condition would normally read as $\|\mathcal{G}\|_\infty \leq 1$. It can be simplified in this case into $\|\mathcal{G}\|_\infty = 1$, since $\|\mathcal{G}\|_\infty \geq |\mathcal{G}(0)| = 1$. The core analytical problem considered in this letter is thus stated as follows:

Problem 1: Given a vehicle model as in (1) with a known τ and a (time-invariant) communication delay that satisfies $\delta \in [0, \bar{\delta}\tau]$ with a known $\bar{\delta}$, find (a lower bound on) the minimum legitimate $h = \bar{h}\tau$ for which $\|\mathcal{G}\|_\infty = 1$.

III. STRING-STABLE CONTROLLER SYNTHESIS

In this section we first investigate the minimum legitimate time headway based on string stability analysis. The main result derived through this investigation is then used to analyze suboptimal controllers from several aspects.

A. Analysis of String Stability

We start our analysis by first investigating the stability requirements for the feedback loop. Since $\bar{h} > 0$, the stability requirement reduces to the analysis of the characteristic polynomial of the feedback loop formed by $\mathcal{P}_0$ and $\mathcal{C}_0$:

$$\mathcal{Q}_0(\bar{s}) = \bar{s}^3 + \bar{s}^2 + \kappa\phi\bar{s} + \kappa. \quad (21)$$

By using Routh-Hurwitz stability criterion, we easily determine the stability requirement for this loop as

$$\phi > 1 \text{ and } \kappa > 0. \quad (22)$$

We now move on to the analysis of $\mathcal{L}_2$ string stability requirement. In preparation for our derivations, we first

recall Euler's identity from complex analysis and express the frequency response function for the delay term as

$$1 - \mathcal{D}(j\omega) = 1 - e^{-j\delta\omega} = \underbrace{1 - \cos(\delta\omega)}_{\varsigma} + j \underbrace{\sin(\delta\omega)}_{\sigma}. \quad (23)$$

The string stability requirement $\|\mathcal{G}\|_{\infty} = 1$ reads as the frequency domain inequality $|\mathcal{G}(j\omega)| \leq 1$. In reference to (18) and (23), this is expressed equivalently as

$$\begin{aligned} & |1 - \mathcal{S}_0(j\omega)[1 - \mathcal{D}(j\omega)]|^2 \\ &= 1 + |\mathcal{S}_0(j\omega)|^2 |1 - \mathcal{D}(j\omega)|^2 - 2\Re\{\mathcal{S}_0(j\omega)[1 - \mathcal{D}(j\omega)]\} \\ &\leq |\mathcal{H}(j\omega)|^{-2} = 1 + \bar{h}^2 \varpi^2, \end{aligned} \quad (24)$$

where $\varpi \triangleq \tau\omega$ is introduced as a scaled frequency variable. Removing one from both sides and multiplying the remaining inequality by $|\mathcal{Q}_0(j\varpi)|^2/\varpi^2$ yields the equivalent condition

$$\begin{aligned} & \bar{h}^2 \cdot \underbrace{(\varpi^6 + (1 - 2\kappa\phi)\varpi^4 + \kappa(\kappa\phi^2 - 2)\varpi^2 + \kappa^2)}_{|\mathcal{Q}_0(j\varpi)|^2} \\ & \geq \underbrace{\kappa(\phi\varpi^2 + 1) \cdot 2\varsigma + 2\kappa(\phi - 1) \cdot \sigma\varpi}_{\kappa(\phi\varpi^2 + 1) \cdot 2\varsigma + 2\kappa(\phi - 1) \cdot \sigma\varpi} \cdot \varpi^{-2} \left(|\mathcal{N}_0(j\omega)|^2 |1 - \mathcal{D}(j\omega)|^2 - 2\Re\{\mathcal{N}_0(j\omega)\mathcal{Q}_0^*(j\omega)[1 - \mathcal{D}(j\omega)]\} \right). \end{aligned} \quad (25)$$

The right hand side of this inequality is equal to zero when $\delta = 0$ ($\Rightarrow \varsigma = \sigma = 0$) thus establishing that $\bar{h}$ can then be chosen arbitrarily close to zero. The condition becomes hard to use exactly in the case of nonzero communication delay due to the trigonometric terms in the right hand side.

To be able to relax (25) into a polynomial inequality, we consider replacing the terms in the right hand side with upper bounds thereof. This is done by employing the well-known trigonometric inequalities

$$2(1 - \cos\theta) \leq \theta^2 \text{ and } \theta \sin\theta \leq \theta^2. \quad (26)$$

The right hand side of (25) is thus bounded from above as

$$\begin{aligned} & \underbrace{\kappa(\phi\varpi^2 + 1)}_{\geq 0} \cdot \underbrace{2\varsigma}_{\leq \bar{\delta}^2\varpi^2} + \underbrace{2\kappa(\phi - 1)}_{\geq 0} \cdot \underbrace{\sigma\varpi}_{\leq \bar{\delta}\varpi^2} \\ & \leq \bar{\delta}^2\kappa\varpi^4 + \bar{\delta}\kappa[\bar{\delta} + 2(\phi - 1)]\varpi^2. \end{aligned} \quad (27)$$

Since the upper bound in (27) converges to zero as $\bar{\delta} \rightarrow 0$, we might be optimistic about relaxing (25) based thereon. The sufficient condition for string stability obtained in this fashion can be reorganized into the form

$$\begin{aligned} & \varphi^3 + (1 - 2\kappa\phi - \bar{h}^{-2}\bar{\delta}^2\kappa\phi)\varphi^2 \\ & + \kappa(\kappa\phi \cdot \phi - 2 - \bar{h}^{-2}\bar{\delta}^2[1 + 2\bar{\delta}^{-1}(\phi - 1)])\varphi \\ & + \kappa^2 \geq 0, \quad \forall \varphi = \varpi^2 \geq 0. \end{aligned} \quad (28)$$

This simply reads as a nonnegativity requirement for a cubic polynomial over the nonnegative real axis.

An analysis of this nonnegativity requirement leads us to some sufficient conditions for string stability expressed in terms of the feedback controller parameters and the communication delay. In reference to [18], we first identify two and only two cases in which (28) can be ensured. Realizing that one of these is an overly complicated condition to work with, we restrict interest to the trivial case in which the coefficients of

φ^2 and φ are required to be nonnegative. A necessary condition for this follows from $\bar{h} \rightarrow \infty \Rightarrow \bar{h}^{-2} \rightarrow 0$ as

$$\frac{2}{\phi} < \psi \triangleq \kappa\phi < \frac{1}{2} \quad (\Rightarrow \phi > 4). \quad (29)$$

This appears as a necessary condition for the nonnegativity of the coefficients even when $\bar{\delta} \rightarrow 0$, while we already know from above that there is actually no string stability requirement on the (ϕ, κ) couple for $\bar{\delta} = 0$. Admitting to this indication of potential conservatism with the hope of deriving some analytical results, we next impose the nonnegativity requirements on the coefficients of φ^2 and φ as follows:

$$\frac{\bar{\delta}^2}{\bar{h}^2} \leq \frac{1}{\psi} - 2, \quad (30)$$

$$\frac{\bar{\delta}^2}{\bar{h}^2} \leq \frac{\psi\phi - 2}{1 + 2\bar{\delta}^{-1}(\phi - 1)}. \quad (31)$$

These are thus identified as sufficient conditions for string stability together with (29). Further analysis leads us to the main result of this letter as a solution to Problem 1:

Theorem 1: The CACC of Figure 1 with an uncertain (yet constant) communication delay $\delta \in [0, \tau\bar{\delta}]$ can be ensured to be string stable with the feedforward and feedback controllers of the form (14) and (15), provided that

$$\bar{h} = \frac{h}{\tau} > \bar{h}_o = \sqrt{2\bar{\delta}(\sqrt{1 + 0.5\bar{\delta}} + 1)}. \quad (32)$$

A suitable feedback controller can then be constructed with

$$\phi = \frac{(2\bar{h}^2 + \bar{\delta}^2)(2\bar{h}^2 - 2\bar{\delta} + \bar{\delta}^2)}{\bar{h}^4 - 4\bar{\delta}\bar{h}^2 - 2\bar{\delta}^3}, \quad (33)$$

$$\kappa = \frac{1}{\phi} \cdot \frac{\bar{h}^2}{2\bar{h}^2 + \bar{\delta}^2}. \quad (34)$$

Proof: For a fixed ϕ , increased values of ψ lead to increased and decreased right hand sides in (30) and (31) respectively. The ψ choice that leads to minimized time headway is hence the one that renders the right hand sides of these equations equal. This brings us to

$$\begin{aligned} & \frac{\psi\phi - 2}{1 + 2\bar{\delta}^{-1}(\phi - 1)} = \frac{1}{\psi} - 2 \Leftrightarrow \\ & \bar{\delta}\phi + 4(\phi - 1)\psi^{-1} - [\bar{\delta} + 2(\phi - 1)]\psi^{-2} = 0. \end{aligned} \quad (35)$$

The best choice of $\psi^{-1} > 0$ is thus obtained as

$$\frac{1}{\psi_o} = \frac{2(\phi - 1)}{\bar{\delta} + 2(\phi - 1)} + \sqrt{\left[\frac{2(\phi - 1)}{\bar{\delta} + 2(\phi - 1)} \right]^2 + \frac{\bar{\delta}\phi}{\bar{\delta} + 2(\phi - 1)}}, \quad (36)$$

which can be shown to be monotone increasing with increasing $\phi > 1$. In reference to (30), we hence conclude that larger choices of ϕ will lead to larger ψ_o^{-1} and thereby to smaller lower bounds on $\bar{h}$. Taking the limit as $\phi \rightarrow \infty$, we derive a necessary condition for the feasibility of (30) with $\psi = \psi_o$ as

$$\frac{\bar{\delta}^2}{\bar{h}^2} < \lim_{\phi \rightarrow \infty} (\psi_o^{-1} - 2) = \sqrt{1 + 0.5\bar{\delta}} - 1, \quad (37)$$

which can be expressed equivalently as in (32). For a fixed $\bar{h} > \bar{h}_o$, we can find $\phi < \infty$ and thereby a legitimate $1/\psi_o$ choice as in (36). Although there would then exist infinitely many controllers that ensure string stability, the most intuitive

and simple choice would be obtained by replacing “ $\leq$ ” in (30) and (31) with “ $=$ ” to find

$$\psi = \frac{\bar{h}^2}{2\bar{h}^2 + \bar{\delta}^2}, \quad (38)$$

$$\phi = \frac{2\bar{h}^2 - 2\bar{\delta} + \bar{\delta}^2}{\bar{h}^2\psi - 2\bar{\delta}}, \quad (39)$$

which lead to (33) and (34). ■

B. Comparative Evaluation of Suboptimal Controllers

Controllers designed based on Theorem 1 will necessarily be suboptimal since (32) is a strict inequality. In other words, we will be able to design controllers with $\bar{h}$ set to values that are arbitrarily close (but not equal) to $\bar{h}_o$. The variation of $\bar{h}_o$ is shown in Figure 2(a), together with two suboptimal $\bar{h}$ choices. The plots are shown in the $\bar{\delta}$ range for which $\bar{h} < 2$, since $\bar{h} = 2$ is known to be achievable with a PD-type feedback-only controller [20].

The parameters of the corresponding suboptimal controllers obtained based on (33) and (34) are shown in Figure 2(b). Interestingly, $\psi = \kappa\phi$ stays in the neighborhood of 0.5 for both controllers, which is the limit value as $\bar{\delta} \rightarrow 0$. As a result the κ values approach zero when getting closer to the optimum time headway.

Considering the two suboptimal controllers, we now analyze the error response as well as the stability robustness and noise sensitivity of the resulting feedback loop.

1) *Error Response to Preceding Vehicle Maneuvers:* As implied by (12), this is determined by the transfer function $\mathcal{M}$, which is obtained for the specific controllers as in (19). The quality of the error response to preceding vehicle maneuvers can be assessed based on a worst-case analysis of $\mathcal{H}_\infty$ and $\mathcal{H}_2$ norms. To eliminate the uncertain delay from this analysis and also achieve some normalization with respect to the vehicle time constant, we recall $\bar{s} = \tau s$ and use the decomposition

$$\mathcal{M}(s) = \tau^2 \cdot \underbrace{\frac{\bar{\delta}\bar{s}}{\mathcal{Q}_0(\bar{s})}}_{\tilde{\mathcal{M}}(\bar{s})} \cdot \underbrace{\frac{1 - e^{-\frac{\bar{\delta}}{s}}}{\bar{\delta}\tau s}}_{\Delta(s)}. \quad (40)$$

It can be shown based on (20) and (26) that $\|\Delta\|_\infty \leq 1$. With $\omega = \tau\omega$, this implies

$$|\mathcal{M}(j\omega)| \leq \tau^2 |\tilde{\mathcal{M}}(j\omega)| \Rightarrow \|\mathcal{M}\|_\infty \leq \tau^2 \|\tilde{\mathcal{M}}\|_\infty. \quad (41)$$

Using this upper bounding, we also infer that

$$\underbrace{\frac{1}{2\pi} \int_{-\infty}^{+\infty} |\mathcal{M}(j\omega)|^2 d\omega}_{\|\mathcal{M}\|_2^2} \leq \tau^4 \cdot \underbrace{\frac{1}{2\pi} \int_{-\infty}^{+\infty} |\tilde{\mathcal{M}}(j\omega)|^2 \cdot \frac{\tau^{-1} d\omega}{d\omega}}_{\tau^3 \|\tilde{\mathcal{M}}\|_2^2}, \quad (42)$$

where $\|\tilde{\mathcal{M}}\|_2$ should be viewed as the $\mathcal{H}_2$ norm of $\tilde{\mathcal{M}}(s)$. We thus conclude that the worst-case response can be assessed (independently of τ) based on $\tilde{\mathcal{M}}(s)$.

Figure 2(c) reveals that the worst-case error response degrades with increasing communication delay, which is quite intuitive. It is perhaps more interesting to observe that approaching optimality also improves the worst-case error response (irrespective of the norm used for evaluation). The worst-case gain from the energy of u_{i-1} to the squared-peak

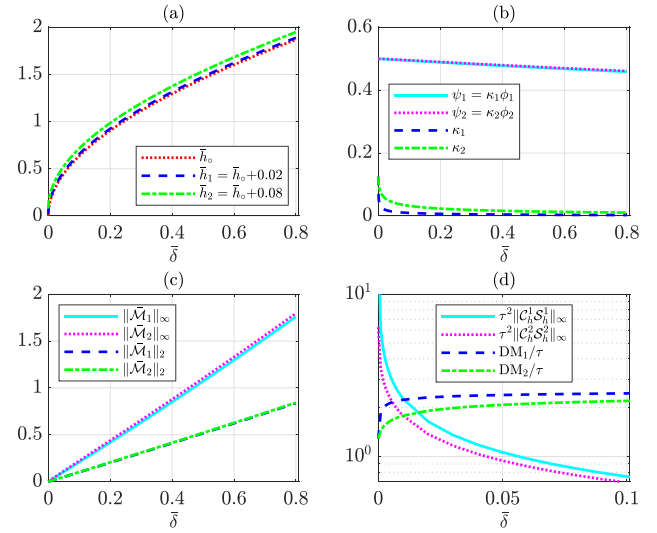


Fig. 2. Comparative evaluation of suboptimal controllers.

of e_i is given by $\|\mathcal{M}\|_2^2$ (see, e.g., [7]). The variation of $\|\tilde{\mathcal{M}}\|_2$ will thus facilitate a quantitative safety analysis (e.g., by assuming the harshest braking scenarios) and thereby a suitable choice of the inter-spacings, g_i , at zero velocity.

2) *Stability Robustness and Noise Sensitivity of the Feedback Loop:* Due to the uncertainty in the vehicle model and the persistent effect of the measurement noise, we restrict the evaluation of the feedback loop to stability robustness and noise sensitivity analysis. Although the sensitivity margin (defined as $1/\|S_h\|_\infty$) can serve as a comprehensive measure for stability robustness, we consider here an analysis of the delay margin (DM) to compensate for neglecting the vehicle delay in our plant model. The DM is defined as the maximum tolerable delay before instability and is computed as the phase margin divided by the gain crossover frequency. The delay margins visualized in Figure 2(d) indicate a good level of stability robustness and more so for the controller that is closer to the optimum. Indeed the delay margin is well above 2τ in the case of large communication delays and approaches 1.3τ in the case of perfect communication.

The noise sensitivity of the feedback loop is determined by $C_h S_h$, whose $\mathcal{H}_\infty$ norm is also displayed in Figure 2(d) (only for small maximum communication delays for better visibility). We identify from this plot a major drawback of getting closer to the optimum time headway as the increase in the noise sensitivity (especially for small communication delays). In view of this, it would be sensible to keep noise sensitivity low by using a desirably large $\bar{\delta}$ value in synthesis.

IV. EXAMPLE SIMULATIONS

To illustrate the platoon behavior, example simulations are performed with the vehicle time constant set to $\tau = 0.5$ s. The communication delay is assumed to be $\bar{\delta} = 0.2\tau$ and the time headway is chosen slightly above the corresponding lower bound as $h = (\bar{h}_o + 0.02)\tau = 0.9253\tau$. We consider four vehicles behind a leader. The simulation results for a scenario in which the leading vehicle brakes harshly after an aggressive acceleration are presented in Figure 3. The plots in the left column relate to the ideal case of no noise or vehicle actuation delay. The plots in the right column are obtained by including some uncorrelated noise in the error measurements (with a

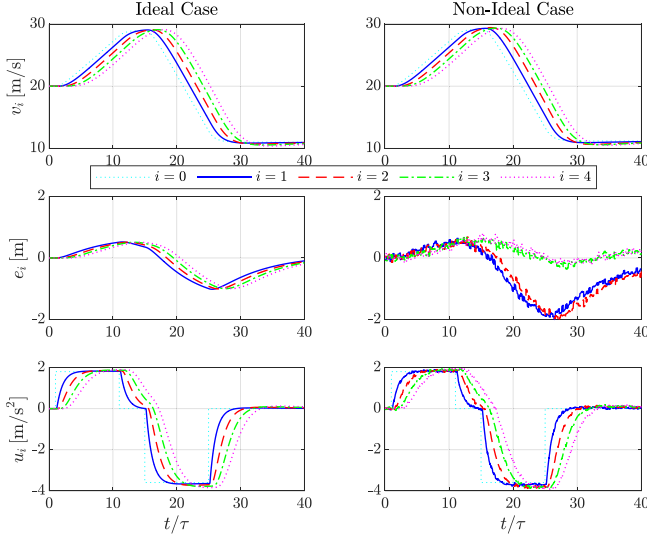


Fig. 3. Simulation results (Communication delays: $\delta_i = 0.2\tau$; Vehicle delays for $t \geq 15\tau$: $v_0 = v_4 = 0.2\tau$, $v_1 = v_3 = 0.3\tau$, $v_2 = 0.4\tau$).

TABLE I
 $\mathcal{L}_2 \rightarrow \mathcal{L}_2$ GAINS FROM SIMULATIONS

	$\ u_1\ _2$	$\ u_2\ _2$	$\ u_3\ _2$	$\ u_4\ _2$
Ideal Case	0.9703	0.9918	0.9966	0.9987
Non-ideal Case	0.9792	1.0016	0.9889	0.9907

level that is visible from the middle plot) and activating some vehicle actuation delays (indicated by v_i) as of the moment of braking by the leader. The $\mathcal{L}_2$ string stability is verified computationally in the noiseless case as in Table I, although we observe a slight overshooting of control inputs especially at the end of braking. This clearly motivates a similar investigation of minimum time headway for $\mathcal{L}_\infty$ string stability. The spacing error is maintained below one meter in the ideal case. There is some visible effect of the measurement noise on the desired acceleration signals, but the $\mathcal{L}_2$ string stability of the control inputs is barely violated even with vehicle delays activated. On the other hand, the negative influence of large vehicle delay is evident especially from the spacing errors of the first and second vehicles, which reach two meters.

V. CONCLUSION

A lower bound is derived for the time headway choices that ensure string stability in one-vehicle look-ahead CACC in the face of communication delay. It is shown that this lower bound can be approached arbitrarily at the cost of an increase in the noise sensitivity. Simulations lead to some optimism about deriving a similar result for heterogeneous platoons with vehicle models that include the vehicle actuation delay as well. To improve the practical applicability, one needs to adopt a framework that is suitable for time-varying delays.

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